



Control Panel UI for Lamina

Feature Requirement Document

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SW

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Document Control

Status	Reviewing
Development Trigger	SW
Target Version	Lamina 1.8-compatible release
Target Release Date	
Approval Sign-Off	Pending

1. Context

Feature Context

The Control Panel provides a centralized Lamina interface for operating and monitoring AXI Panel Handler hardware, including Rail Width, Conveyor Belt, Outer Barriers, Panel Clamps, and Safety Level Sensors.

The feature replaces fragmented hardware interaction with coordinated controls, confirmed hardware-state feedback, software interlocks, and operation status.

Panel Handler control requirement → multiple hardware subsystems affected → Lamina must provide a safe and consistent operator interface.

2. User Personas

2.1 Primary User

R&D / Machine Integration Engineer

The primary user operates the Control Panel during AXI machine development, integration, testing, and troubleshooting.

The user needs to:

- directly operate Panel Handler hardware;
- view current hardware states;
- adjust supported positions and offsets;
- identify when an operation is unavailable; and
- identify hardware or communication failures.

2.2 Secondary User

QA / Validation Engineer

The secondary user verifies:

- hardware control behaviour;
- UI-to-hardware state synchronization;
- safety and operation interlocks;
- input validation;
- startup and shutdown behaviour;
- error handling; and
- recovery behaviour.

The feature may also be used by service personnel for hardware diagnosis where direct Panel Handler control and state visibility are required.

3. Functional Requirements

3.1 Inputs

The Control Panel shall provide operator inputs for the following functions:

3.1.1 Rail Width Control

The UI shall allow the operator to:

1. Enter a target Rail Width.
2. Select the engineering unit used to enter and display the Rail Width.
3. Apply the entered Rail Width.
4. Home the Rail Width axis.
5. View and edit the Rail Width home offset.
6. Select the engineering unit used for the home offset.
7. Apply an updated home offset.
8. Initiate Rail Width reinitialization when an unsafe limit condition requires recovery.

The requested Rail Width shall be validated against the minimum and maximum position limits provided by the active machine configuration before movement is requested. Invalid, empty, non-numeric, below-minimum, and above-maximum inputs shall be rejected before a movement command is sent.

Rail Width adjustment and homing shall be unavailable while the Conveyor Belt is running. If Rail Width homing or adjustment is requested while the rail axis is disabled, the system shall enable the axis before performing the operation.

3.1.2 Conveyor Belt

The operator shall be able to:

1. select conveyor direction;
2. start conveyor movement; and
3. stop conveyor movement.

The conveyor direction shall not be changeable while the conveyor is running. The conveyor must be stopped before another direction can be selected. The Conveyor Belt shall only be allowed to start while the Panel Clamps are open.

3.1.3 Outer Barriers

The operator shall be able to independently:

1. open the left Outer Barrier;

2. close the left Outer Barrier;
3. open the right Outer Barrier; and
4. close the right Outer Barrier.

The displayed barrier state shall be synchronized with the corresponding hardware sensor state.

3.1.4 Panel Clamps

The operator shall be able to:

1. open the Panel Clamps; and
2. close the Panel Clamps.

The UI shall distinguish between:

- fully closed;
- open; and
- partially closed

A partially closed condition shall not be presented as fully clamped. Panel Clamp operation shall be unavailable while the Conveyor Belt is running.

3.1.5 Safety Level Sensors

The operator shall be able to control the left and right Safety Level Sensors independently.

The available predefined positions are:

Position	Nominal Position
HOME (equivalent to M30L on V810Qx)	Home position
M4	10,000,000 nm
M8L	19,000,000 nm
M15L	41,000,000 nm
M22L	67,000,000 nm
M30L	90,000,000 nm

The operator shall be able to:

1. select a predefined position independently for the left and right Safety Level Sensor;
2. move the selected sensor to the requested position;
3. initiate the Safety Level Sensor Home operation;
4. view and edit the home offset for each sensor; and
5. select the engineering unit used for each editable offset.

A command targeting one Safety Level Sensor shall not cause movement of the opposite sensor, except for the Home command, which homes both Safety Level Sensor axes.

3.2 Observable Processing Behaviour

3.2.1 Initialization Feedback

When the Control Panel is initialized, the interface shall display a loading state while the required hardware subsystems are initialized and their initial states are retrieved. Normal hardware controls shall not become available until the required initialization has completed successfully.

After initialization, the Control Panel shall retrieve the applicable:

- offsets;
- axis positions;
- conveyor state;
- clamp state;
- barrier state;
- Safety Level Sensor states; and
- Rail Width limit states.

3.2.2 Hardware Operation Feedback

During individual hardware operations, the affected control shall provide operation feedback until the operation succeeds or fails.

This applies to:

- Rail Width movement;
- Rail Width homing;
- offset updates;
- Safety Level Sensor movement;
- Safety Level Sensor homing;
- conveyor operations;
- barrier operations; and
- clamp operations.

3.2.3 Confirmed State

The UI shall distinguish between operator-requested values and hardware-confirmed values. Displayed hardware values shall be refreshed from confirmed hardware state after applicable operations, and requested values shall not be presented as successfully applied if the operation fails.

Only one conflicting hardware operation shall be allowed at a time. Controls that could conflict with an operation in progress shall remain unavailable until the operation completes or fails.

3.3 Outputs & Display

The Control Panel shall be divided into five primary hardware sections:

1. Rail Width
2. Conveyor Belt
3. Outer Barriers
4. Panel Clamps
5. Safety Level Sensors

Each section shall display the current relevant hardware state together with the controls required to operate that hardware.

3.3.1 Rail Width Display

The Rail Width section shall display:

- current/applied Rail Width;
- editable target Rail Width;
- selected engineering unit;
- home offset;
- positive limit safe/not safe state;
- negative limit safe/not safe state;
- control availability; and
- recovery/reinitialization status when an unsafe Rail Width condition is detected.

The applied Rail Width shall represent a successfully applied or confirmed hardware value rather than unconfirmed edit text.

3.3.2 Conveyor Belt Display

The Conveyor Belt section shall display:

- whether the belt is running or stopped; and
- the selected/active direction.

3.3.3 Outer Barrier Display

The Outer Barrier section shall independently display the current left and right barrier states.

3.3.4 Panel Clamp Display

The Panel Clamp section shall indicate whether the clamps are:

- fully closed/clamped;
- open; or

- partially closed where applicable.

3.3.5 Safety Level Sensor Display

The Safety Level Sensor section shall independently display for the left and right sides:

- applied predefined position;
- actual position after hardware read-back;
- home-offset value;
- safety input state; and
- home-sensor state.

A hardware position that has not yet been successfully retrieved shall not be displayed as a confirmed physical position.

3.4 Units & Display Conventions

3.4.1 Position Units

Rail Width and Safety Level Sensor position values use nanometres for backend communication. The UI shall perform the required engineering-unit conversion for operator input and display.

3.4.2 Home Offset Units

Home offsets shall be presented using the supported UI engineering-unit conventions and converted to the backend-required representation before being submitted. Editing state and successfully applied state shall remain separate so that editing a new value does not falsely indicate that the hardware has already moved. The selected engineering unit shall remain visible while a home-offset value is being edited.

4. User Interface (UI) & User Experience (UX)

4.1 Desktop Integration

The feature shall be integrated into the existing Lamina desktop application as the Control Panel / Panel Handler interface. The page shall use the existing Lamina visual language and reusable controls while presenting the five Panel Handler hardware sections in a consistent layout. The feature shall not require a separate standalone application.

4.2 Workflow & Navigation

4.2.1 Normal Workflow

The normal operator workflow shall be:

1. Open the Control Panel.
2. Allow the required Panel Handler-related subsystems to initialize.
3. Wait while the current hardware configuration and hardware states are retrieved.
4. Review the displayed Rail Width, conveyor, barriers, clamps, Safety Level Sensors, and safety/limit states.
5. Select the required hardware section.
6. Enter or select the required command or target value.
7. Execute the operation.
8. Observe operation/loading feedback.
9. Hardware state is read back where applicable.
10. Wait for completion.
11. View the refreshed hardware state.

After successful initialization, the Control Panel shall retrieve the current offsets, axis positions, conveyor state, clamp state, barrier state, safety-sensor status, and Rail Width limit state before normal interaction is enabled.

4.2.2 Rail Width Recovery Workflow

If Rail Width enters a recovery-required state:

1. Normal Rail Width movement controls shall be disabled.
2. The interface shall indicate that reinitialization is required.
3. The operator shall initiate recovery/reinitialization.
4. The Control Panel shall display progress while recovery occurs.
5. Rail Width position, enabled state, and limit status shall be refreshed.
6. Normal Rail Width controls shall become available only after the unsafe condition is cleared.

4.3 Interactivity

The Control Panel shall use direct controls appropriate to each hardware section, including:

- editable numeric fields;
- engineering-unit selection;
- buttons for movement, homing, opening, closing, starting, stopping, applying, and reinitializing;
- segmented controls for mutually exclusive selections such as conveyor direction;
- predefined Safety Level Sensor position selection; and
- status indicators for hardware and limit states.

Controls shall be unavailable when:

- initialization is incomplete;
- the required axis is unavailable;
- another conflicting hardware operation is active;
- an applicable software interlock is active; or
- Rail Width recovery is required.

Zooming, panning, rotating, or scientific plot manipulation is not applicable to this feature.

4.4 Feedback & Progress

The UI shall provide two levels of operation feedback:

- a page-level/loading state during subsystem initialization; and
- localized operation feedback for individual hardware commands.

The UI shall remain responsive while hardware operations are running. A failed operation shall terminate its loading state and present an error rather than leaving the interface indefinitely in a busy state. A successful operation shall refresh the applicable hardware state before presenting the resulting value as confirmed.

4.5 State & Mode Indication

The Control Panel shall clearly distinguish between operator-requested state and confirmed hardware state.

Hardware status indicators shall include, where applicable:

- running / stopped;
- open / closed;
- fully clamped / open / partially clamped;
- safe / unsafe;

- home / away from home;
- Rail Width limit safe / unsafe;
- enabled / disabled;
- loading / operation in progress; and
- reinitialization required.

Left and right Safety Level Sensor safe and home states shall be presented independently. Rail Width home state and the Safe/Unsafe status of its positive and negative limits shall also be presented independently.

5. Non-Functional Requirements

5.1 Display Precision

Position and offset values shall use the existing Lamina engineering-unit conversion and formatting behaviour. Internal hardware values shall not be rounded in a manner that changes the value sent to the backend. Display formatting may use the precision appropriate to the selected engineering unit.

No new physical movement accuracy or homing-repeatability requirement is introduced by this FRD. These remain governed by the existing machine and Motion subsystem specifications.

5.2 UI Responsiveness

Hardware communication shall not cause observable blocking of the Lamina UI. The user shall remain able to observe operation progress and error feedback while hardware operations execute.

Digital IO and Panel Handler monitoring shall provide regular status updates, with the implemented monitoring interval nominally 500 ms. The 500 ms interval represents UI/status monitoring frequency and is not a safety-system response-time guarantee.

5.3 Hardware & Platform Constraints

The feature shall operate within the existing Lamina desktop environment and supported AXI Panel Handler hardware configuration. No additional computing hardware, machine hardware, sensors, actuators, controllers, or network dependency is introduced by the feature. The feature depends on the Panel Handler axes and hardware functions available in the active machine configuration.

5.3.1 Key Assumptions

Assumption	Impact if Incorrect
Required Panel Handler hardware is present in the active configuration	Corresponding UI function cannot be safely enabled
Backend reports axis availability correctly	Unsupported motion could otherwise be requested
Hardware sensor states correctly represent barrier, clamp, safety and limit conditions	UI status may not represent the physical machine
Compatible frontend/backend/service versions are deployed together	Operations may be unavailable or fail due to interface incompatibility

Existing hardware safety mechanisms remain authoritative	UI software interlocks must not be treated as replacement safety protection
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The Control Panel provides secondary software protection; existing hardware and backend safety mechanisms remain authoritative.

5.4 Offline Capability

The Control Panel shall operate within the existing local Lamina and AXI service environment. Physical-machine operation uses the connected AXI hardware environment, while simulation mode provides simulated machine states to the UI without requiring a physical machine connection. No new Internet or cloud dependency is introduced by this feature.

6. Edge Cases & Error Handling

6.1 Back-End Error Presentation

When an operation fails, the Control Panel shall:

- stop the associated loading state;
- avoid presenting the requested value as successfully applied;
- retain the previous confirmed state where appropriate;
- display an operation error to the user; and
- prevent further commands when the failure leaves the hardware in a state requiring recovery.

Hardware, communication, monitoring, startup, and shutdown failures shall be surfaced through the existing Lamina error presentation mechanism. The UI shall not display a false success state when the backend operation fails.

6.2 Connection & Communication Failures

If a required backend service or hardware subsystem becomes unavailable:

- the current operation shall be aborted;
- the affected functionality shall not report success;
- the user shall be informed of the failure;
- controls requiring the unavailable subsystem shall remain unavailable where appropriate; and
- monitoring/listener state shall be cleaned up when required.

If initialization cannot be completed, normal Control Panel operations shall not become available.

6.3 Invalid User Input

The Control Panel shall reject:

- empty Rail Width input;
- non-numeric Rail Width input;
- Rail Width below the configured minimum;
- Rail Width above the configured maximum;
- empty home-offset input;
- invalid/non-numeric home-offset input; and
- operations targeting an unavailable Panel Handler axis.

Invalid input shall not overwrite the previously confirmed applied value. Configured position limits shall be obtained from the backend rather than duplicated as hardcoded frontend values.

6.4 Hardware Fault Display

When a hardware fault or software interlock prevents an operation, the Control Panel shall prevent the affected command and communicate the reason to the operator. This includes Rail Width unsafe-limit conditions, Rail Width operations while the Conveyor Belt is running, attempts to start the Conveyor Belt while the Panel Clamps are closed, attempts to operate the Panel Clamps while the Conveyor Belt is running , and attempts to change conveyor direction while the belt is running. Where Rail Width recovery is required, the UI shall indicate that reinitialization is necessary and shall keep normal Rail Width controls unavailable until recovery succeeds and the unsafe condition is cleared.

Safety Level Sensor safe and home states shall be displayed independently for the left and right sides, and the UI shall reflect the confirmed state returned from the corresponding hardware inputs.

7. Backward / Version Compatibility

7.1 UI Compatibility Requirements

The Control Panel is intended for the corresponding Lamina 1.8-compatible backend/service version set. The frontend shall use the active backend configuration to determine available Panel Handler axes instead of assuming that Rail Width and both Safety Level Sensor axes are always present. A hardware command shall not be issued for an axis that is not available in the active configuration. The feature does not require modification of existing physical hardware layouts.

8. Testing, Validation, & Reproducibility

8.1 UI Test Cases

ID	Requirement	Test	Expected Result
TC-01	3.2.1	Open Control Panel	Loading shown; current hardware states retrieved; controls enabled only after successful initialization
TC-02	3.1.1	Apply valid Rail Width	Movement completes and confirmed width is displayed
TC-03	3.1.1, 6.3	Enter invalid/out-of-range Rail Width	Command rejected; previous applied value retained
TC-04	3.1.1	Home Rail Width	Homing completes and confirmed state is refreshed
TC-05	3.1.1, 6.4	Run conveyor and request Rail Width movement/homing	Rail command rejected
TC-06	3.1.2, 6.4	Change direction while conveyor runs	Direction remains unchanged
TC-07	3.1.4, 6.4	Operate clamps while conveyor runs	Clamp command rejected
TC-08	3.1.2, 6.4	Start conveyor with Panel Clamps partially or fully closed	Conveyor command rejected
TC-09	3.1.3	Operate left/right barriers independently	Only requested barrier changes
TC-10	3.1.4, 3.3.4	Produce intermediate clamp sensor state	UI does not report fully clamped
TC-11	3.1.5	Home Safety Level Sensors	Both Safety Level Sensor axes home and their confirmed states are refreshed
TC-12	3.1.5, 3.3.5	Test M4/M8L/M15L/M22L/M30L	Correct selected axis moves; opposite axis is unaffected; confirmed state refreshes
TC-13	3.1.1, 4.2.2, 6.4	Trigger unsafe Rail Width limit and perform recovery	Normal controls disabled and reinitialization required; controls restored only after successful recovery and cleared limit state
TC-14	3.2.2, 3.2.3, 6.1	Force hardware operation failure	Loading terminates; error is presented; no false applied-value update
TC-15	4.3, 6.3, 7.1	Request unavailable axis	No movement command reaches hardware
TC-16	4.4, 5.2	Execute hardware operation	UI remains responsive and shows operation feedback
TC-17	3.1.1, 3.1.5, 3.4.2, 6.3	Edit and apply Rail Width and Safety Level Sensor home offsets	Valid values are applied; invalid values are rejected; displayed applied value updates only after successful completion

TC-18	6.1, 6.2	Cause hardware/service communication failure	Operation does not report success; error is presented; affected controls remain unavailable where appropriate
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8.2 Acceptance Criteria

Approval requires all applicable test cases to pass. The following measurable conditions shall be met:

Acceptance Measure	Pass Criterion
Unintended opposite Safety Level Sensor movement	0 occurrences
Rejected interlock commands reaching backend	0 commands
Invalid or out-of-range input resulting in hardware command	0 commands
Unavailable-axis commands reaching native movement	0 commands
False applied-value updates after failed operation	0 occurrences
Failed operations remaining indefinitely in loading state	0 occurrences
Rail Width recovery completing while an unsafe limit remains active	0 occurrences
Observable UI-thread blocking during hardware operation	0 occurrences
Shutdown hangs caused by monitoring cleanup	0 occurrences in validation cycle

Repeated lifecycle validation shall complete successfully on the supported physical-machine configuration, with no duplicate monitoring, deadlock, shutdown hang, or stale hardware access. Simulation-mode validation shall be limited to functionality supported by the simulator.

8.3 Usability Validation

Usability shall be validated by representative engineering/QA users operating the Control Panel on the approved AXI machine.

Without direct backend/API interaction, the user shall be able to:

- identify each hardware-control section;
- determine the current state of each controlled component;
- operate each supported hardware function;
- distinguish editable/requested values from confirmed hardware values;
- identify when an operation is in progress;
- understand why an interlocked operation is unavailable;

- recognize an unsafe Rail Width condition and initiate reinitialization; and
- operate the left and right Safety Level Sensors as defined by the supported control behaviour.

Pass criterion: All required Control Panel workflows shall be completed without direct backend/API intervention or ambiguity between requested and confirmed hardware state.

Downstream Impact Flags

Downstream impact	Y/N	(What changes, for whom)
Mfg procedure / calibration change	N	No new manufacturing or hardware calibration procedure is introduced.
Field service / diagnostics change	Y	Control Panel provides additional centralized hardware control, status visibility, limit indication, and Rail Width recovery behaviour.
Training / documentation change	Y	Users operating the Control Panel require documentation of the five hardware sections, software interlocks, editable offsets, Safety Level Sensor controls, and Rail Width recovery behaviour.
Deployment risk (rollback, compatibility)	Y	Control Panel functionality depends on compatible frontend and backend/service components.

Appendix — Control Panel UI Reference

1. Control Panel Overview

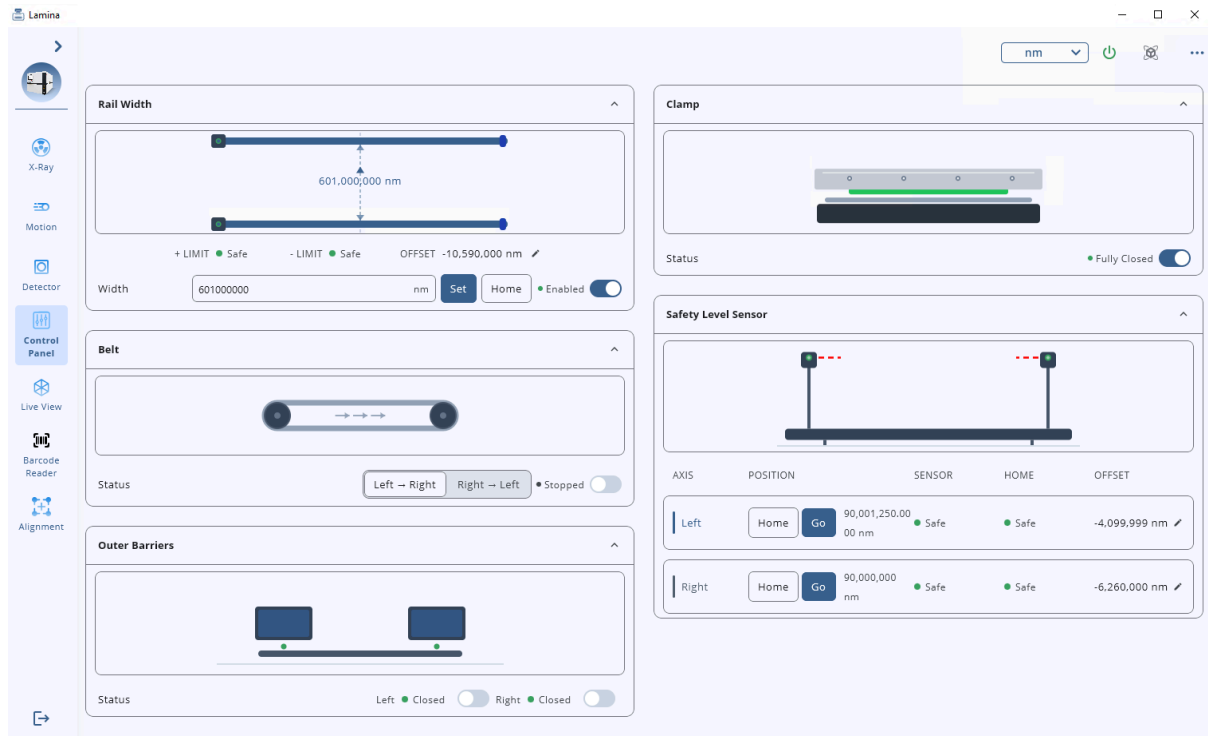


Figure 1: Control Panel overview showing the five hardware-control sections: Rail Width, Conveyor Belt, Outer Barriers, Panel Clamps, and Safety Level Sensors.

2. Rail Width

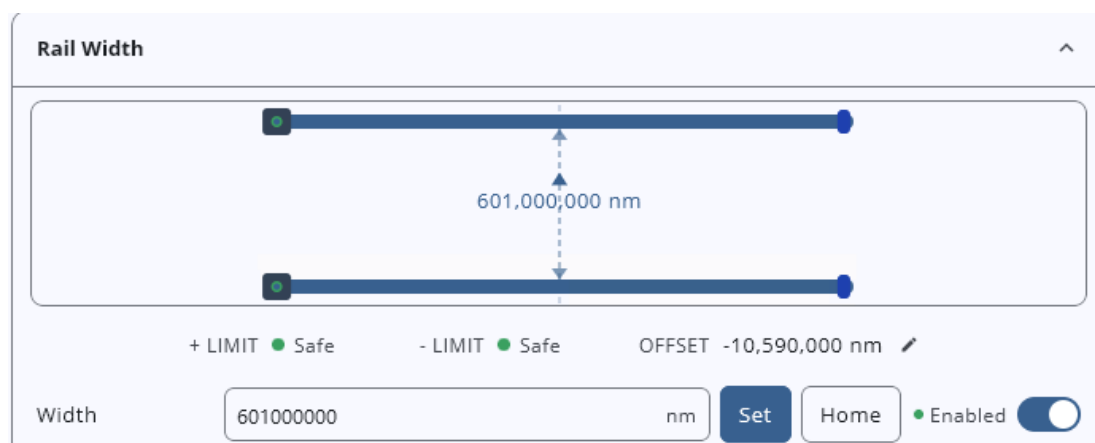


Figure 2: Rail Width section after initialization, showing the axis enabled and the machine-reported initial Rail Width of 601,000,000 nm at the Home position.

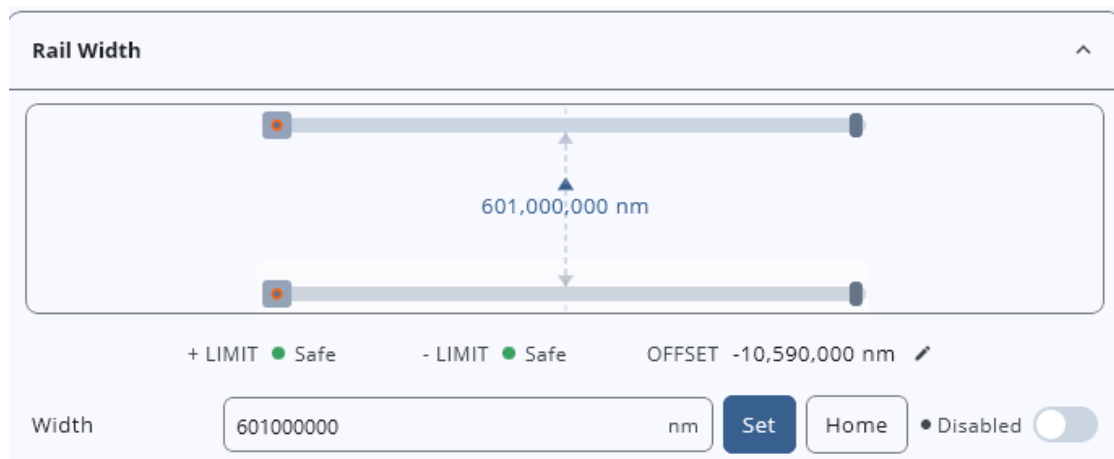


Figure 3: Rail Width section showing the disabled-axis state and corresponding UI indication.

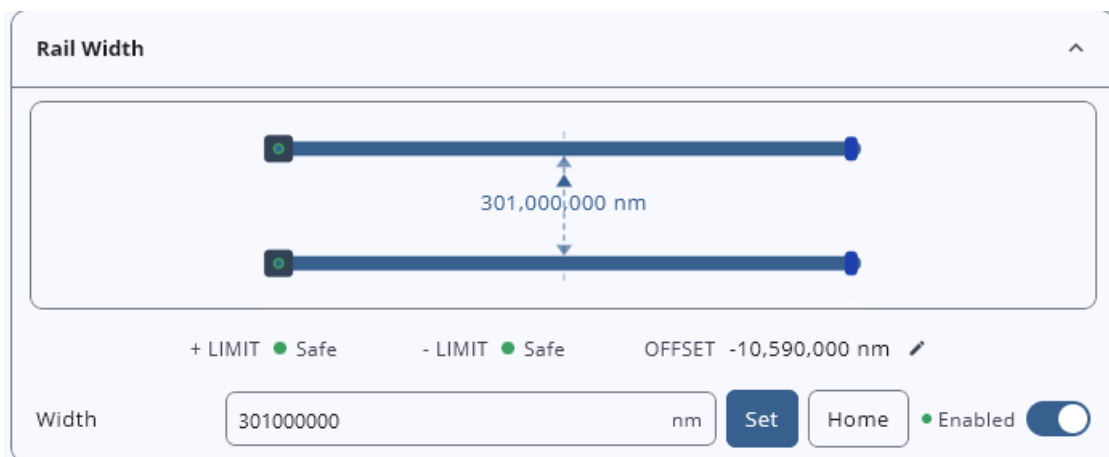


Figure 4: Rail Width after applying a target of 301,000,000 nm, showing the updated diagram, applied value, and confirmed actual-position read-back after physical movement.

3. Conveyor Belt

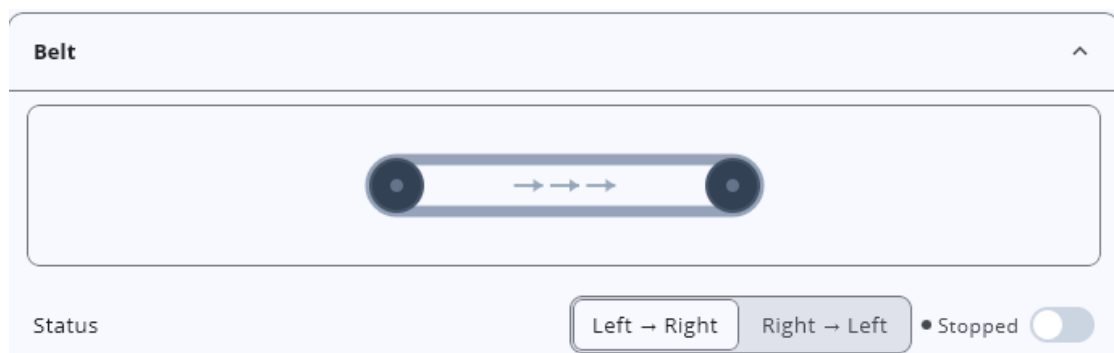


Figure 5: Conveyor Belt initial state after initialization, showing the belt stopped with Left-to-Right direction selected.

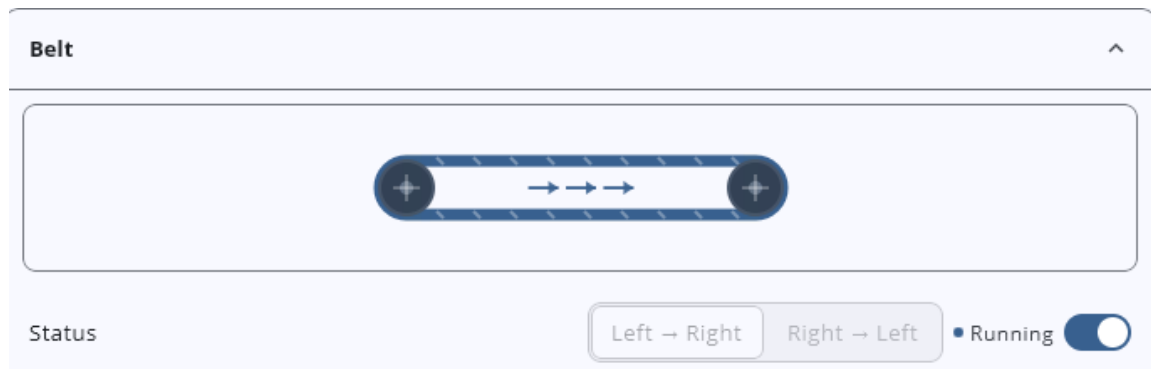


Figure 6: Conveyor Belt running in the selected Left-to-Right direction, showing the corresponding running-state UI feedback.

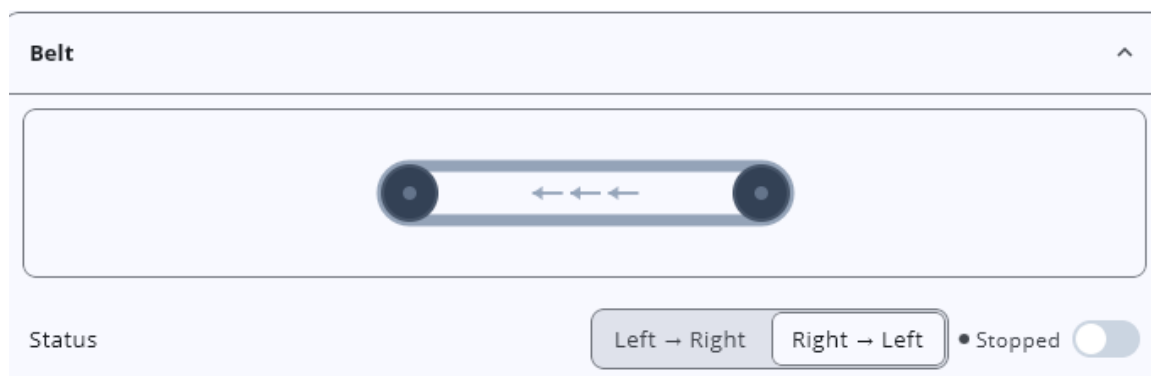


Figure 7: Conveyor Belt direction changed to Right-to-Left after the belt was stopped, demonstrating that direction selection is unavailable during movement.

4. Outer Barriers

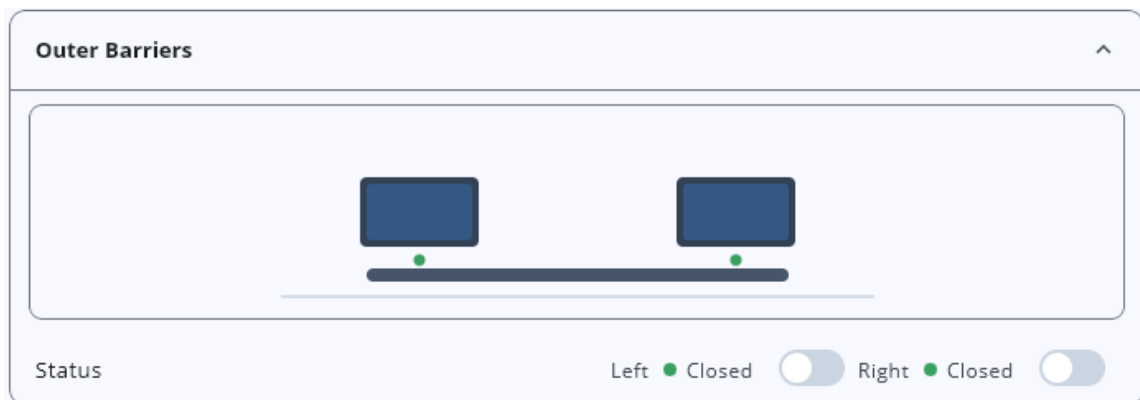


Figure 8: Outer Barrier initial state after initialization, showing both left and right barriers closed.

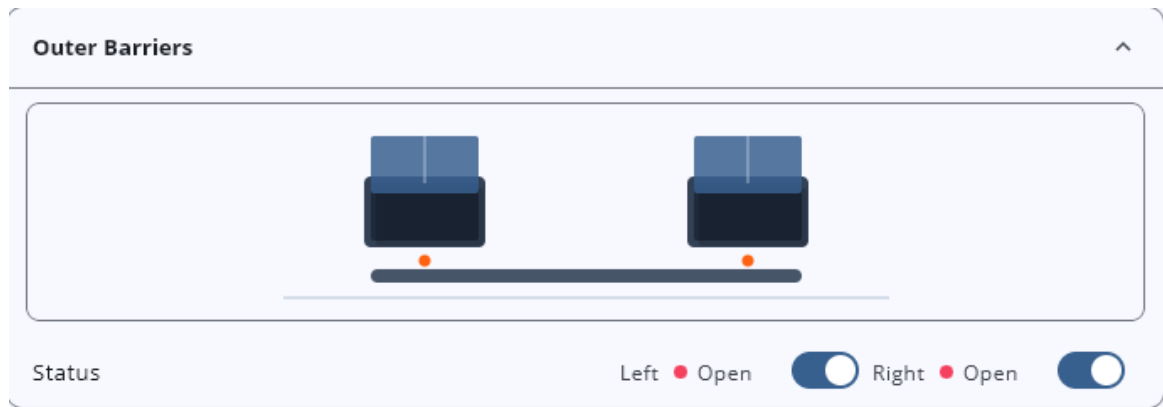


Figure 9: Outer Barrier section with the left barrier opened independently and the UI updated to reflect the corresponding hardware state.

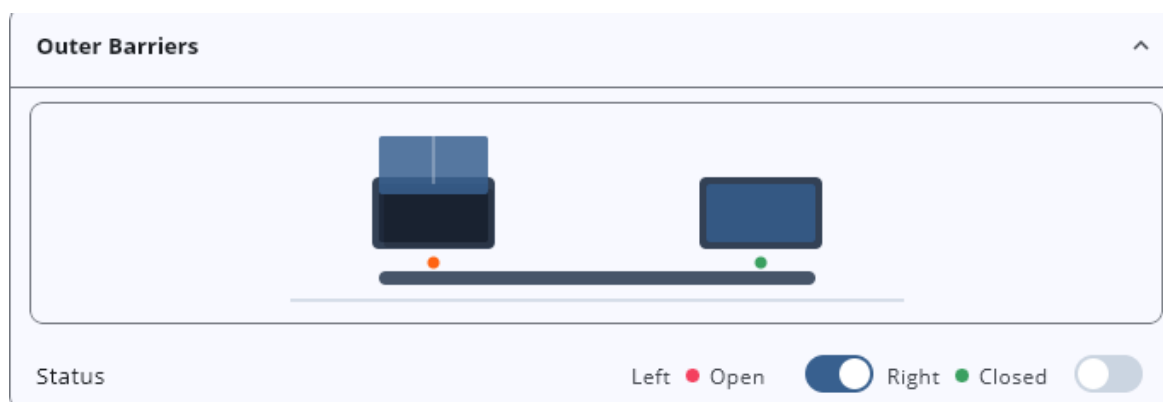


Figure 10: Outer Barrier section showing both left and right barriers open and independently reflected in the UI.

5. Panel Clamps

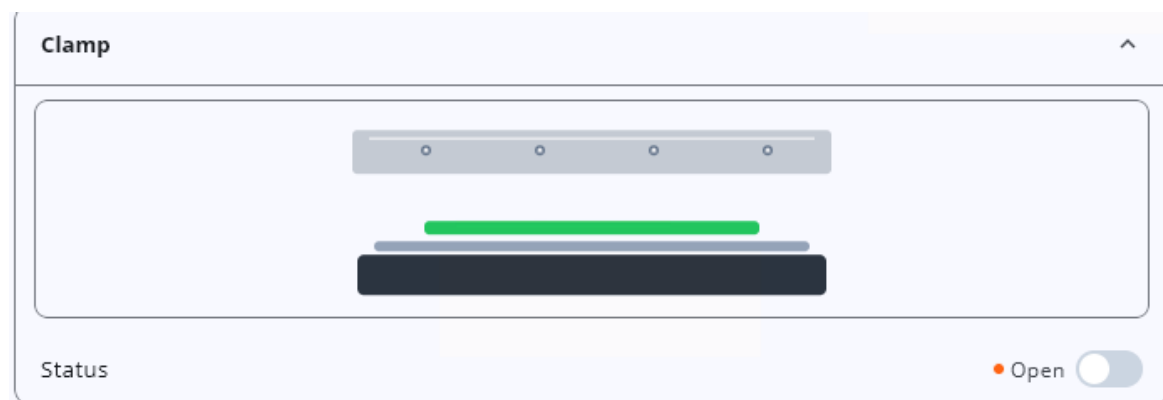


Figure 11: Panel Clamp initial state showing the clamps confirmed closed after initialization.

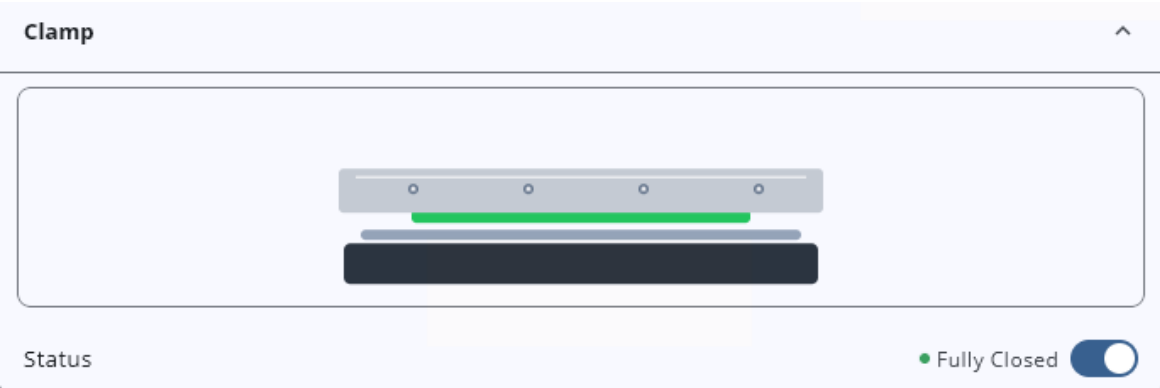


Figure 12: Panel Clamp section after opening the clamps, showing the updated open-state indication.

6. Safety Level Sensors

Safety Level Sensor

The diagram shows a safety level sensor assembly with two vertical posts and a horizontal base. Each post has a green dot at the top, and a red dashed line extends from each dot. The horizontal base is dark grey.

AXIS	POSITION	SENSOR	HOME	OFFSET	
Left	<button>Home</button> <button>Go</button>	90,001,250.00 00 nm	● Safe	● Safe	-4,099,999 nm
Right	<button>Home</button> <button>Go</button>	90,000,000 nm	● Safe	● Safe	-6,260,000 nm

Figure 13: Safety Level Sensor initial state showing both left and right sensors at the Home position after initialization.

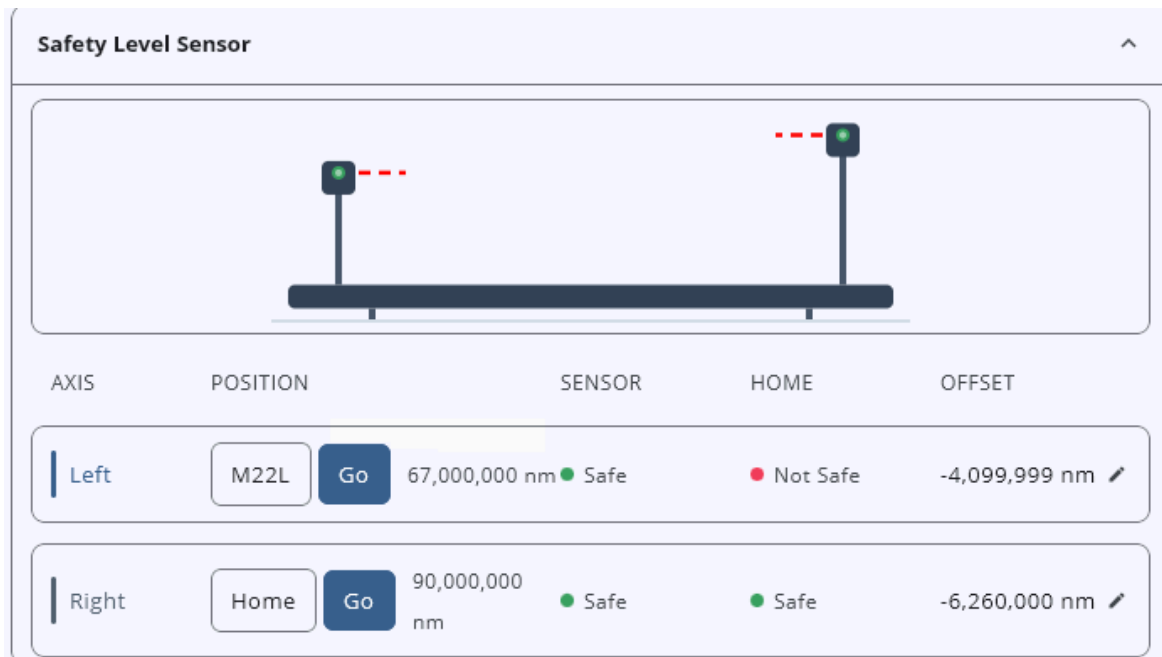


Figure 14: Left Safety Level Sensor moved to M22L, showing the updated predefined position, actual-position read-back, and corresponding UI diagram.

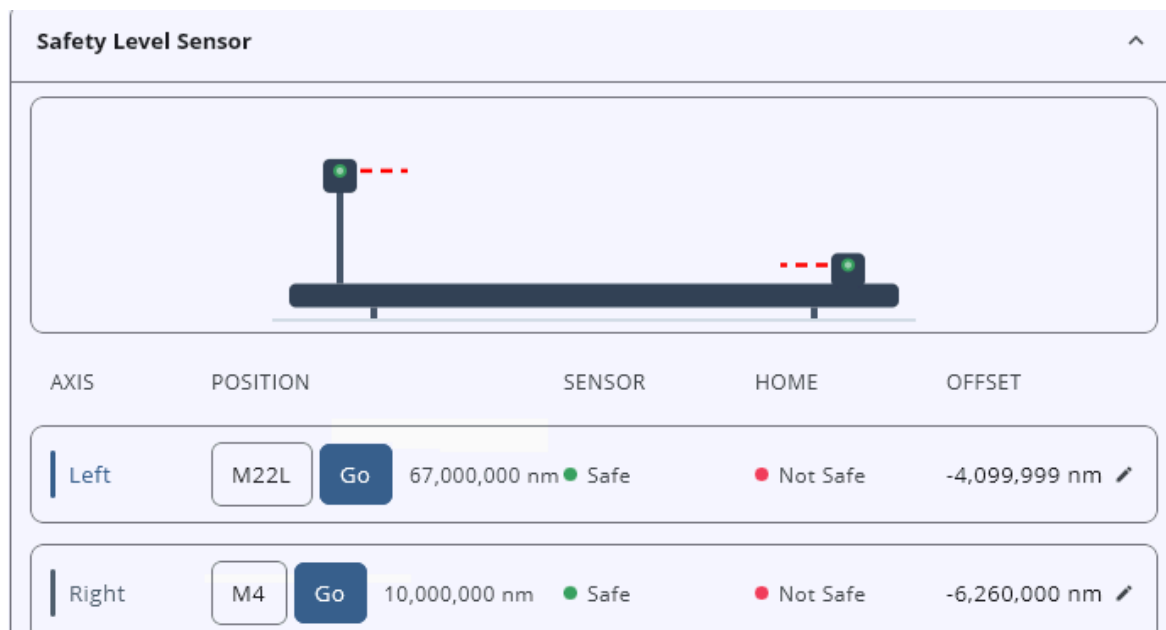


Figure 15: Right Safety Level Sensor moved independently to M4 while retaining the previously selected left-side state, demonstrating independent position control.

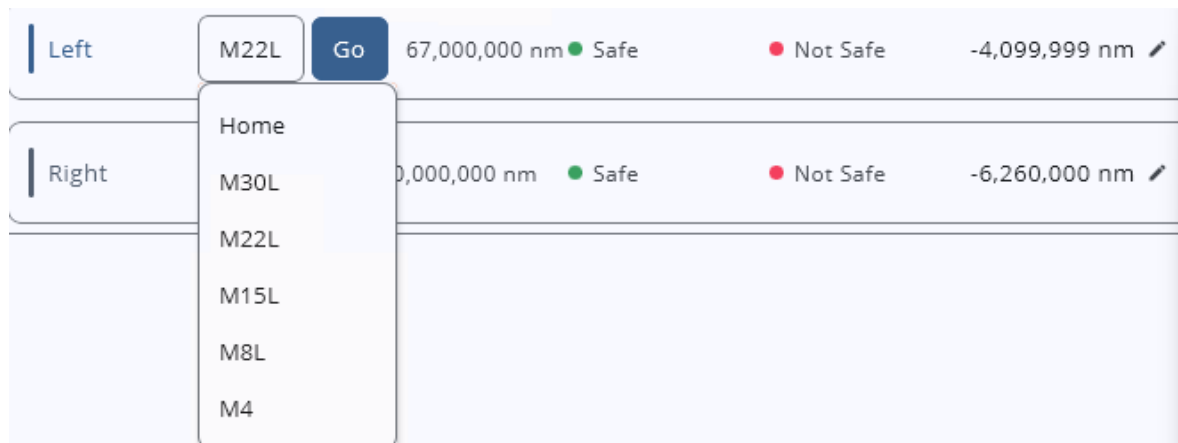


Figure 16: Safety Level Sensor position selector showing the available predefined positions: Home, M4, M8L, M15L, M22L, and M30L. Home and M30L correspond to the same physical position on V810Qx.

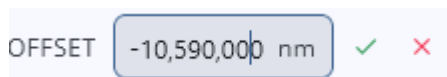


Figure 17: Home-offset edit mode showing the editable value, confirmation and cancellation controls, and the selected engineering unit remaining visible during editing.

7. Editing and Engineering Units

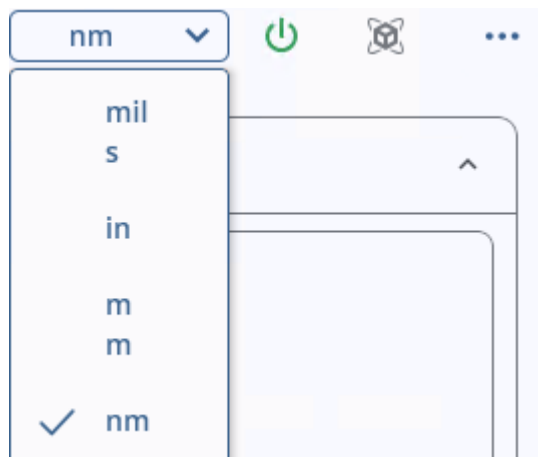


Figure 18: Engineering-unit selector showing supported display/input units including mils, inches, millimetres, and nanometres; nanometres are used for backend machine communication.

8. Application Appearance and Localization

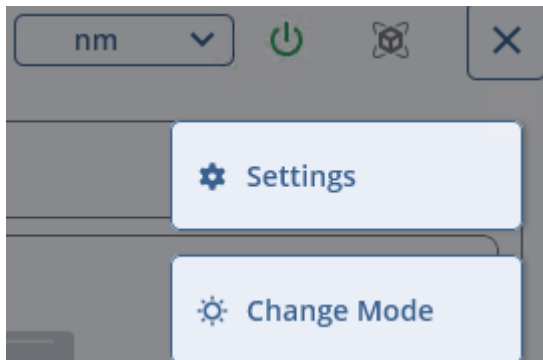


Figure 19: Control Panel application menu showing access to Settings and Change Mode options.

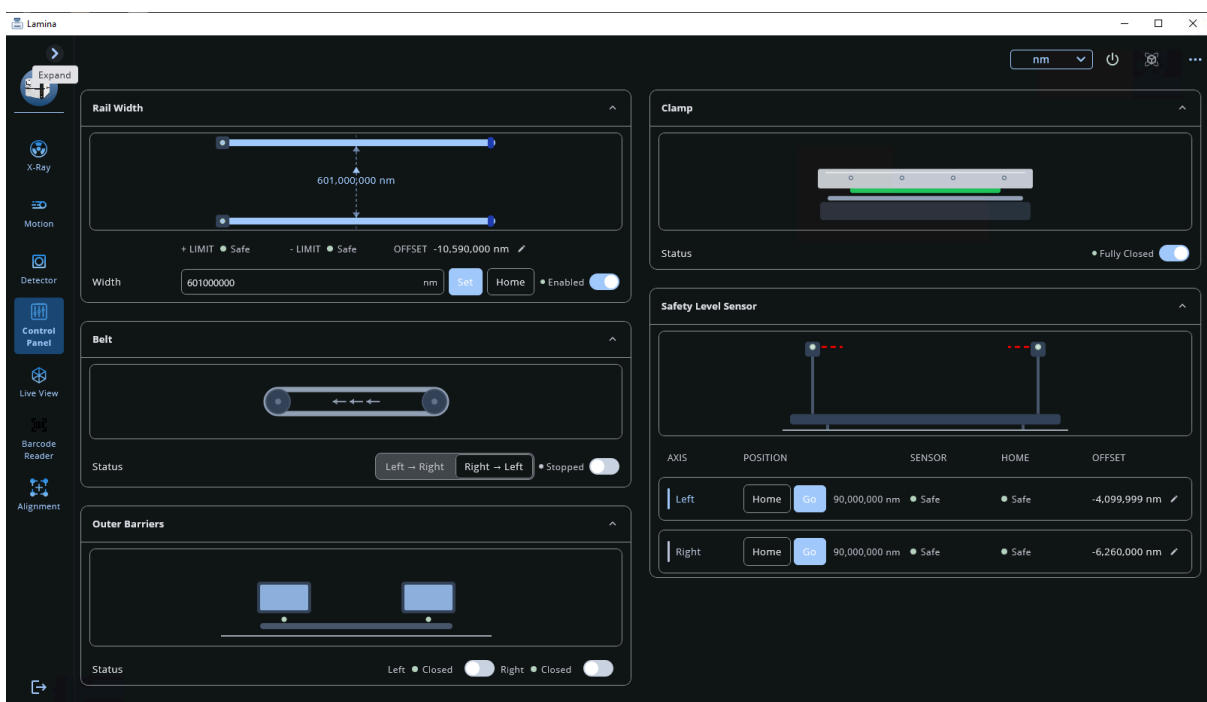


Figure 20: Control Panel displayed using the Lamina dark appearance mode, with Control Panel components adapting to the selected application theme.

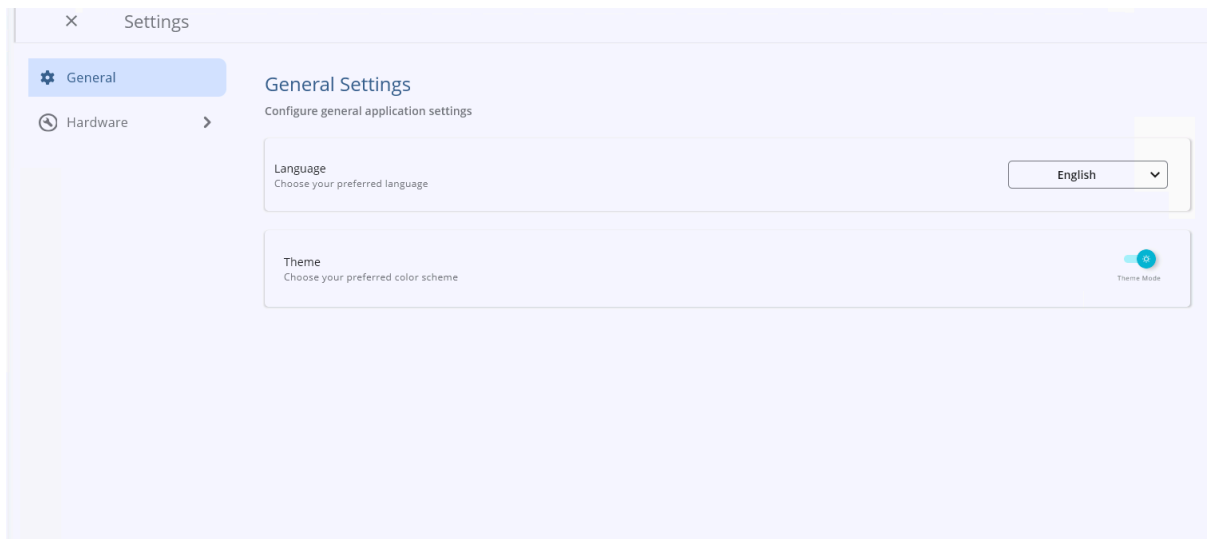


Figure 21: Application language setting showing the available language configuration for the Lamina interface.

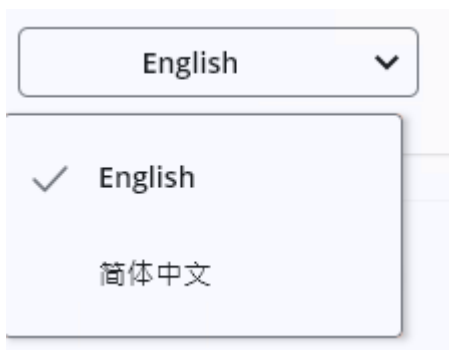


Figure 22: Language-selection dropdown showing Chinese (Mandarin) as a selectable application language.

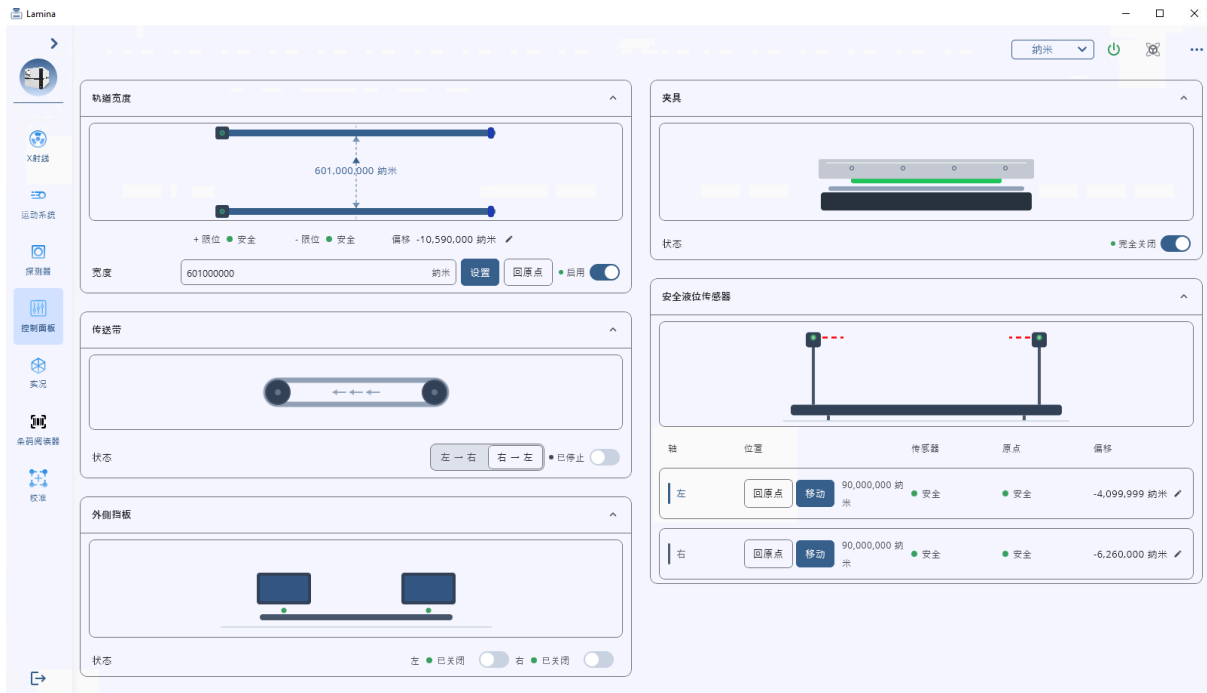


Figure 23: Control Panel displayed in Chinese (Mandarin), demonstrating localization of UI labels, controls, status messages, and loading feedback.

9. Interlock and Recovery Feedback

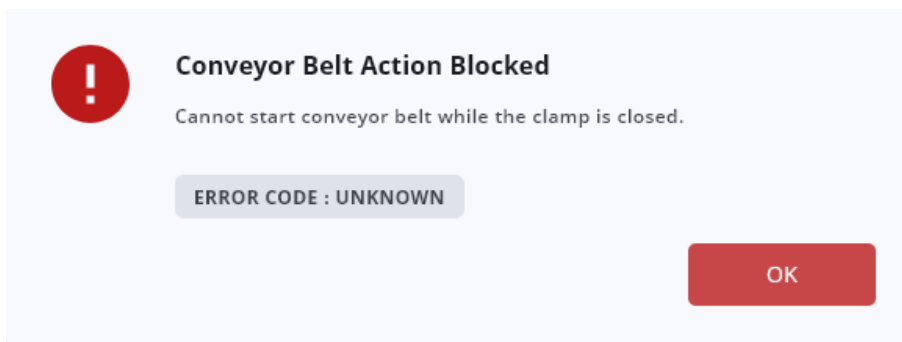


Figure 24: Conveyor Belt action-blocked message displayed when conveyor movement is prevented by the Panel Clamp interlock.

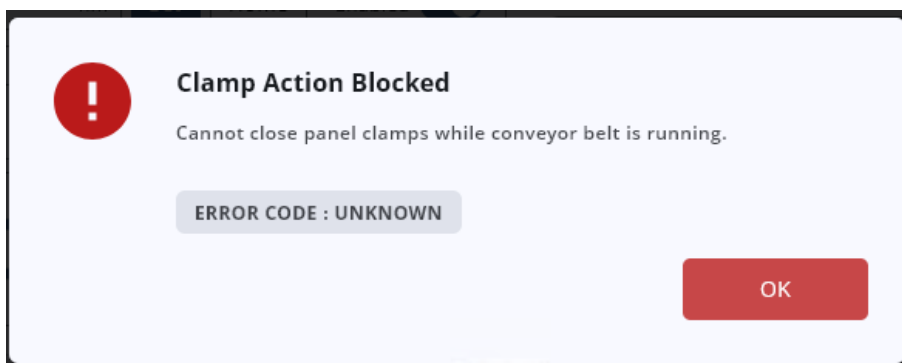


Figure 25: Panel Clamp action-blocked message displayed when a clamp operation conflicts with the running Conveyor Belt state.

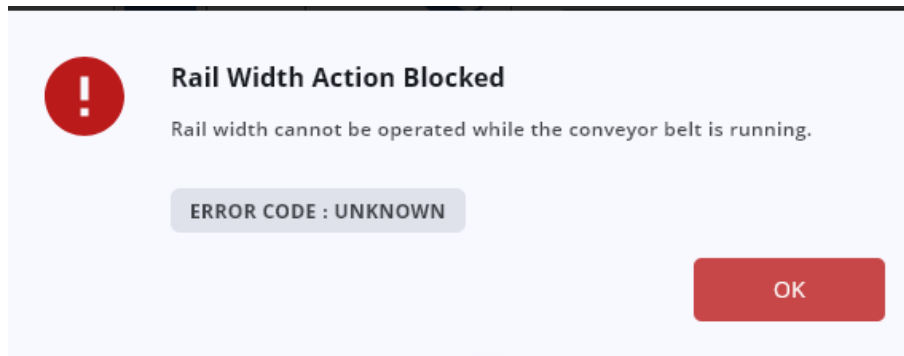


Figure 26: Rail Width action-blocked message displayed when Rail Width operation is attempted while the Conveyor Belt is running.

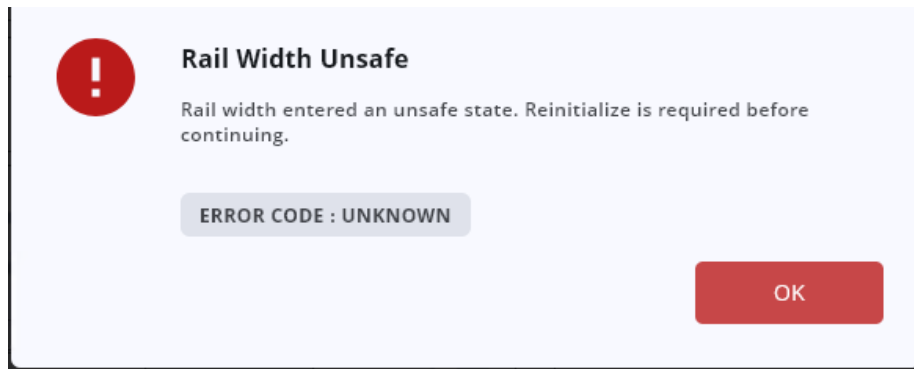


Figure 27: Rail Width unsafe-state indication showing that reinitialization is required before further Rail Width operation and that the lifecycle control changes to the initialization action.

10. Module Lifecycle Feedback

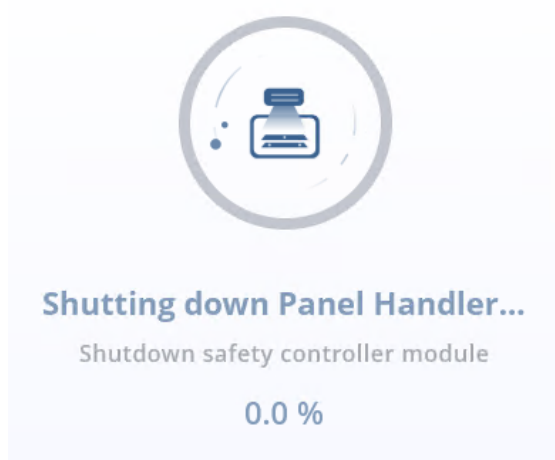


Figure 28: Control Panel during subsystem shutdown, showing Panel Handler and Safety Controller module shutdown progress and corresponding lifecycle status feedback.

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